

GOKUL SRINATH

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SUMMARY

Senior Software Engineer with 5.5+ years building production-grade AI systems, ML pipelines, and distributed backends. Shipped a 6-agent LLM platform (LangGraph) serving 5,000+ employees and a payroll fraud detection pipeline (85–90% precision) on 10K+ monthly transactions. Delivered 90%+ runtime reductions across multiple systems via distributed, serverless-first architecture on AWS. Hands-on across the full AI stack – RAG pipelines, agentic workflows, data preparation, and LLM evaluation.

SKILLS

Languages	JavaScript, TypeScript, Python, Java
Frontend	React.js, Redux, TanStack Query, React Hook Form, TailwindCSS
Backend	Node.js (TypeScript), Express.js, FastAPI
AI / ML	LangGraph, RAG, DSPy, RAGAS, Isolation Forest, RuleFit, SMOTE, Vector Search, BM25, Hybrid Search, Cross-Encoder Reranking
Cloud	AWS (Lambda, SQS, Step Functions, API Gateway, DynamoDB), GCP (Cloud Functions, Pub/Sub, Log Router), Azure, Serverless
Databases	SQL, MongoDB, Redis, Qdrant, pgvector
Architecture	Distributed Systems, Event-Driven Architecture, Microservices, JWT, OAuth

EXPERIENCE

Senior Software Engineer

May 2025 – Present

4iApps Solutions – Payday (*AI-powered HR & Payroll Platform*)

Tech Stack: Python, Node.js (TypeScript), React.js, SQL, Redis, AWS (Lambda, SQS, Step Functions, DynamoDB), LangGraph, DSPy, RAGAS, Isolation Forest, RuleFit, SMOTE, Qdrant, BM25, TailwindCSS

- Architect and maintain a reusable multi-agent LLM platform using LangGraph with 6 domain-specific agents (Payroll, Payslip, HR Attendance, JD Writer, Resume Parser, Fraud Detection) serving 5,000+ employees; design a modular Query Interpreter narrowing 200+ candidate APIs to 15 per request with Redis-backed 12-turn stateful memory
- Implement a hybrid RAG pipeline combining semantic chunking, vector search, BM25, RRF, and cross-encoder reranking; eliminate prompt brittleness via DSPy’s declarative optimization compiled from curated QA pairs, cutting prompt iteration from days to hours; measure quality via RAGAS-based LLM-judge evaluation scripts
- Develop a production payroll fraud detection pipeline using Isolation Forest and RuleFit, achieving 85–90% precision and 80–85% recall on 10K+ monthly transactions; engineer domain-specific features (salary progression deltas, grade transition signals) and apply SMOTE to handle a highly imbalanced dataset
- Build an agentic SQL + AWS pipeline completing anomaly detection on 10K transactions in under 5 minutes post-payroll, routing ~50 flagged cases per cycle to HR via dedicated review screens with LLM-generated natural language explanations – replacing a previously manual, multi-day process
- Slash batch processing from 1–2 hours to under 90 seconds (95%+ reduction) via parallelized event-driven queues; cut costing pipeline from 2 hours to 6–8 minutes (93% reduction) using AWS Step Functions with dynamic Lambda concurrency; reduce infrastructure compute spend 40% by migrating always-on ECS Fargate to event-driven Lambda
- Lead technical delivery for a 5-engineer team, setting architecture direction and maintaining 3–4 production releases per sprint; compress 2-week sprint planning to under 3 hours using AI-assisted tooling (Cursor + Figma MCP)

Software Engineer

2020 – May 2025

Virtusa

Tech Stack: Node.js, Python, Serverless Framework, GCP (Cloud Functions, Pub/Sub, Log Router), AWS (CloudFormation, S3, IAM, DaaS), Azure (Logic Apps, Runbooks), Qdrant

- Contributed to a centralized cloud administration and unified asset platform for JP Morgan, consolidating management of 1,000+ infrastructure assets (DaaS, S3, IAM) into a single interface; assisted in automating bulk resource provisioning via CloudFormation templates, reducing multi-day manual workflows to on-demand execution
- Assisted in developing a serverless GCP vulnerability detection system using Cloud Functions, Pub/Sub, and Log Router that monitors 30+ cloud accounts from a single control plane, automatically flagging 3–4 newly created vulnerable resources per day
- Contributed to an internal incident intelligence tool for the cloud cost management platform – wrote Python scripts to parse and normalize ServiceNow incident exports and chunk internal runbook markdown into structured passages; applied metadata tagging (cloud provider, service, severity) before ingestion into a Qdrant vector store; the RAG pipeline enabled on-call engineers to query past incidents and retrieve runbook recommendations via an internal web UI, reducing average incident resolution lookup time

PROJECTS

AI Code Review Agent

2025

github.com/gokul-srinath/ai-dev-assistant

Tech Stack: Python, FastAPI, AWS Bedrock (Amazon Nova), Qdrant, Tree-sitter, BM25, Cross-Encoder, GitHub Webhooks, Docker

- Built an AI-powered code review agent triggered via GitHub webhooks that posts structured PR feedback using AWS Bedrock (Amazon Nova); implemented AST-based chunking via Tree-sitter to split files by functions and classes, preserving semantic boundaries over arbitrary token splits
- Designed a hybrid retrieval pipeline combining Qdrant vector search and BM25, reranked with a cross-encoder; built a dual-index strategy with isolated PR-scoped collections for unmerged changes and a separate main index for production code, preventing context pollution across concurrent PRs
- Integrated PRD validation by embedding requirement docs, retrieving relevant sections per PR diff, and validating actual file contents against a strict met/not-met checklist

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering

2017 – 2021

Sri Shakthi Institute of Engineering & Technology

CGPA: 8.0

KEY ACHIEVEMENTS

- Slashed batch processing 95%+ (1–2 hrs → <90 sec) and costing pipeline 93% (2 hrs → 6–8 min) via distributed, event-driven architecture on AWS
- Shipped 6-agent LLM platform serving 5,000+ employees – hybrid RAG, 200+ API intent routing, 12-turn stateful memory
- Delivered payroll fraud ML pipeline (Isolation Forest + RuleFit): 85–90% precision, 80–85% recall on 10K+ monthly transactions, replacing manual HR review
- Cut infrastructure costs 40% via serverless migration; compressed sprint planning 95% with AI-assisted tooling
- Automated cyber crisis migration of 10,000+ users with zero data loss: 3+ days manual → 2-hour fully automated execution